Fiscal Year:	FY 2007	**Task Last Updated:**	FY 12/06/2007
PI Name:	Ingber, Donald M.D., Ph.D.		
Project Title:	Mechanotransduction Through Integrins		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	donald.ingber@childrens.harvard.edu	**Fax:**	FY 617-730-0230
PI Organization Type:	UNIVERSITY	**Phone:**	617-919-2223
Organization Name:	Children's Hospital / Harvard Medical School		
PI Address 1:	300 Longwood Avenue		
PI Address 2:	Karp Family Research Laboratory, 11th floor, Room 128		
PI Web Page:	web1.tch.harvard.edu/research/ingber/homepage.htm		
City:	Boston	**State:**	MA
Zip Code:	02115	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	07/01/2004	**End Date:**	06/30/2008
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 6/30/2008 per A. Chu/ARC (6/07) 8/06--funding reinstated thru 6/30/007, per J. Cohen (HQ)		
Key Personnel Changes/Previous PI:	none		
COI Name (Institution):			
Grant/Contract No.:	NNA04CC96G		
Performance Goal No.:			
Performance Goal Text:			

	1. To compare the effects of mechanical stresses of different intensity, frequency and direction on focal adhesion assembly, cell strengthening and G protein activation 2. To analyze the Rho signaling mechanism responsible for cell strengthening behavior 3. To identify the molecules that mediate transmembrane mechanochemical coupling between integrins and G proteins within the focal adhesion
Task Description:	Over the past year, we showed that intracellular microtubules do bear large-scale compressive loads from a variety of physiological forces, but their buckling wavelength is reduced significantly because of mechanical coupling to the surrounding elastic cytoskeleton. We quantitatively explained this behavior, and showed that this coupling dramatically increases the compressive forces that microtubules can sustain, suggesting they can make a more significant structural contribution to the mechanical behavior of the cell than previously thought possible.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Elucidation of the mechanism by which integrins mediate cellular mechanotransduction should help us to understand the molecular basis of the cellular response to gravity as well as many other forms of mechanosensation and tissue regulation. Through the proposed research program, we hope to identify specific molecular mediators of mechanochemical transduction as well as novel mechanical signaling pathways in mammalian cells. These molecules may serve as potential targets for future modes of therapeutic intervention in diseases, such as atherosclerosis, hypertension, and osteoporosis, which are characterized by abnormal force transduction by cells. Further insight into mechanobiology also may impact on design of artificial tissues and medical devices for tissue repair and transplantation.
Task Progress:	Mechanical forces transferred via transmembrane integrin receptors from the extracellular matrix (ECM) to the cytoskeleton control cell shape and initiate multiple biochemical signaling cascades within cells that regulate critical cell behaviors, including growth, motility, differentiation and apoptosis. However, the molecular mechanism by which cells convert mechanical signals into a biochemical response still remains poorly understood, and elucidating this mechanism is the major goal of this project. In last year's Progress Report, we described progress we made in pursuit of Specific Aims 1 and 2. During the past year, we focused on Specific Aim 3. Work during previous funding periods showed that the heterotrimeric G protein subunit Gas is recruited to focal adhesions that form within minutes following ß1 integrin clustering and activation induced by cell binding to magnetic microbeads (4.5 um diameter) coated with integrin ligands. We also showed that application of mechanical stresses to these microbeads using applied magnetic fields leads to rapid activation of heterotrimeric G proteins at focal adhesions, stimulation of adenylyl cyclase, and activation of downstream cAMP signaling events that result in gene transcription, whereas application of the same force to other membrane receptors had little effect. Over the past year, we focused our efforts on further analyzing the role of Gas in mechanosensitive cAMP signaling. We found that rapid activation of cAMP signaling is induced when endothelial cells adherent to flexible ECM-coated substrates are exposed to cyclic mechanical strain as measured by quantitation of nuclear CREB phosphorylation. To determine if Gas activation is required for cAMP signaling in response to application of physiological forces to cell adhesion receptors, we utilized siRNA to knock down Gas expression. While nuclear phospho-CREB staining increased significantly in control cells relative to non-stimulated cells within 5 minutes after strain application, this response was significantly inhibited in cells treated with Gas siRNA. Moreover, chemical inhibition of Gas activity with mellittin also significantly reduced strain-induced nuclear CREB phosphorylation. Thus, Gas appears to be required for cAMP signaling induced by mechanical forces transferred across ECM adhesions in these endothelial cells. Accurate understanding of cellular mechanotransduction through integrins and G proteins requires knowledge of how heterotrimeric G proteins associate with focal adhesions. Focal adhesions are large, complex and dynamic structures and the full complement of focal adhesion proteins are not known. Therefore, to determine the adaptor proteins bridging integrins to G proteins, we made a series of integrin mutants designed to disrupt large portions of the focal adhesion. Over the past 12 months, we genetically engineered a series of six consecutive mutations covering the transmembrane and cytoplasmic domains of the ß1 integrin based on known interactions from the literature. These mutations are all made in chimeric constructs where the extracellular domain of integrin has been replaced with carbonic anhydrase IV (CAIV) epitope allowing us to selectively bind and study our mutants in the presence of wildtype integrins. We have now generated two sets of integrin mutants, one with the ß1 transmembrane domain and another where the transmembrane domain has been replaced with the low density lipoprotein (LDL) transmembrane domain. Importantly, in preliminary studies using these mutants, we found that the cytoplasmic region of ß1 integrin is sufficient to support focal adhesion formation as determined by cell spreading and fluorescence microscopic analysis. Our preliminary studies indicate that, as expected, the various cytoplasmic mutants have differential spreading efficiencies. In addition, some have decreased focal adhesion strength in response to pulling through focal adhesions formed on CAIV antibody-coated beads despite their ability to spread on CAIV coated substrates indicating that the focal adhesion is partially disrupted. Therefore, these chimeras should be extremely useful tools for the dissection of mechanosensitive signal transduction through integrins and G proteins.
Bibliography Type:	Description: (Last Updated: 02/25/2021)
Articles in Peer-reviewed Journals	Brangwynne CP, MacKintosh FC, Kumar S, Geisse NA, Talbot J, Mahadevan L, Parker KK, Ingber DE, Weitz DA. "Microtubules can bear enhanced compressive loads in living cells because of lateral reinforcement." J Cell Biol. 2006 Jun 5;173(5):733-41. PMID: 16754957 , Jun-2006
Articles in Peer-reviewed Journals	Parker KK, Ingber DE. "Extracellular matrix, mechanotransduction and structural hierarchies in heart tissue engineering." Philos Trans R Soc Lond B Biol Sci. 2007 Aug 29;362(1484):1267-79. Review. PMID: 17588874 , Aug-2007
Articles in Peer-reviewed Journals	Stamenovic D, Rosenblatt N, Montoya-Zavala M, Matthews BD, Hu S, Suki B, Wang N, Ingber DE. "Rheological behavior of living cells is timescale-dependent." Biophys J. 2007 Oct 15;93(8):L39-41. Review. PMID: 17693464 , Oct-2007

Articles in Peer-reviewed Journals	Sharp ZD, Mancini MG, Hinojos CA, Dai F, Berno V, Szafran AT, Smith KP, Lele TP, Ingber DE, Mancini MA. "Estrogen-receptor-alpha exchange and chromatin dynamics are ligand- and domain-dependent." J Cell Sci. 2006 Oct 1;119(Pt 19):4101-16. Erratum in: J Cell Sci. 2006 Oct 15;119(Pt 20):4365. Lele, Tanmay T [corrected to Lele, Tanmay P]. PubMed PMID: 16968748 , Oct-2006
Articles in Peer-reviewed Journals	Guo Y, Eichler GS, Feng Y, Ingber DE, Huang S. "Towards a holistic, yet gene-centered analysis of gene expression profiles: a case study of human lung cancers." J Biomed Biotechnol. 2006;2006(5):69141. PubMed PMID: 17489018; PubMed Central PMCID: PMC1698264 , Nov-2006
Articles in Peer-reviewed Journals	Ingber DE, Levin M. "What lies at the interface of regenerative medicine and developmental biology?" Development. 2007 Jul;134(14):2541-7. PubMed PMID: 17553905 , Jul-2007
Articles in Peer-reviewed Journals	Ghosh K, Ingber DE. "Micromechanical control of cell and tissue development: implications for tissue engineering." Adv Drug Deliv Rev. 2007 Nov 10;59(13):1306-18. Review. PubMed PMID: 17920155 , Nov-2007
Books/Book Chapters	Matthews BD, Thodeti CK, Ingber DE. "Integrin-dependent force transmission to mechanosensitive ion channels." in "Mechanosensitive Ion Channels." Ed. O.P. Hammill. Amsterdam ; Boston : Academic Press, c2007. p. 59-85., Apr-2007
Books/Book Chapters	Lele TP, Sero JE, Matthews BD, Kumar S, Xia S, Montoya-Zavala M, Polte T, Overby D, Wang N, Ingber DE. "Tools to study cell mechanics and mechanotransduction." in "Methods in Cell Biology." Ed. Y. Wang, D. Discher. Methods Cell Biol. 2007;83:443-72. PMID: 17613320, Apr-2007